Day 3: Introduction to IP Addressing

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Course: Diploma in Cyber Security

Topic: Day 3: Introduction to IP Addressing

Introduction

An IP address identifies a device on a network. IPv4 uses 32-bit addresses, while IPv6 uses 128-bit addresses. Subnetting divides a network into smaller networks and helps use IP addresses in an organised way.

CIDR notation, such as /24 or /27, tells us how many bits are used for the network part of an IP address.

Task 1

IPv4	**IPv6**
32-bit address	128-bit address
Written as four decimal parts, e.g. 192.168.1.10	Written using hexadecimal groups, e.g. 2001:db8::10
Much smaller address space	Very large address space
Uses ARP for IPv4 address-to-MAC resolution	Uses Neighbor Discovery instead of ARP

Task 2

Device	IP Address	Subnet Mask	Default Gateway
Laptop	192.168.10.10	255.255.255.0	192.168.10.1
Phone	192.168.10.11	255.255.255.0	192.168.10.1
Smart TV	192.168.10.12	255.255.255.0	192.168.10.1
Printer	192.168.10.13	255.255.255.0	192.168.10.1

These are example static addresses. They should not be used if another device on the real network already uses them.

Task 3

Subnet	Network Address	Broadcast	Usable Host Range
1	192.168.1.0	192.168.1.31	192.168.1.1 – 192.168.1.30
2	192.168.1.32	192.168.1.63	192.168.1.33 – 192.168.1.62
3	192.168.1.64	192.168.1.95	192.168.1.65 – 192.168.1.94

Task 4

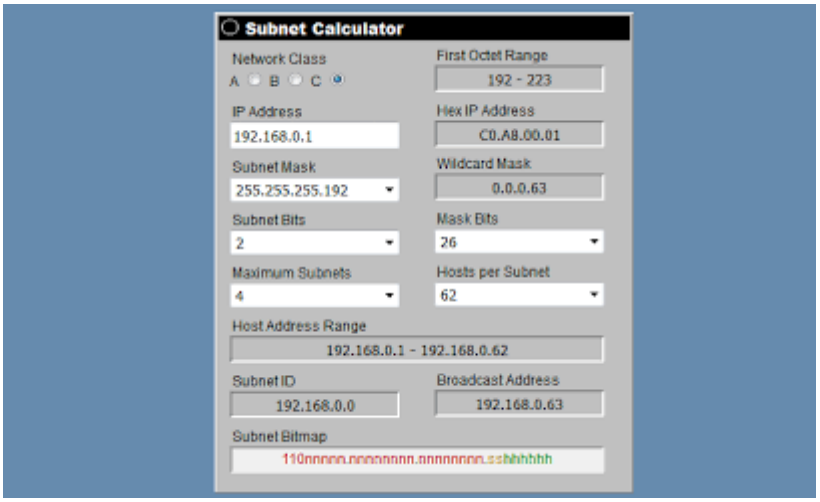
The IP address is 10.0.5.17 with mask 255.255.255.0.
The mask is /24.

Network address: 10.0.5.0
Broadcast address: 10.0.5.255
Usable host range: 10.0.5.1 to 10.0.5.254

Task 5

PRACTICAL

Open the subnet calculator website supplied by your trainer or use the given subnet-calculator.com address. Enter 10.0.5.17 and 255.255.255.0.



Screenshot: Add a screenshot of your actual calculator result.

Theory-based expected values:
Network: 10.0.5.0
Broadcast: 10.0.5.255
Usable hosts: 10.0.5.1–10.0.5.254

Write: “My manual answer matched the calculator output.” only after checking your actual result.

Conclusion

IP addressing is important because devices need addresses to communicate. Subnetting helps divide networks into smaller sections. The examples above show how CIDR notation can be used to find subnet ranges and usable hosts.

Quick Revision

- IPv4 = 32 bits.
- IPv6 = 128 bits.
- /24 = 255.255.255.0.
- /27 has 32 addresses and 30 usable hosts per subnet.
- Network and broadcast addresses are not normally assigned to end devices.